

Exploring RTs per phase (non-linear modelling)

This vignette contains a full breakdown of the exploratory non-linear modelling conducted to analyse the evolution of participants' response times over the course of the trial phases. As stated in the manuscript:

In addition to the total response times per trial usually analysed in VIIE studies, we collected response time data for each trial phase (three premises and the conclusion). We explored whether the VIIE could be specific to certain phases, or whether it was a difference in the dynamics of reasoning across trial phases rather than an overall difference in speed, by modelling response times throughout a trial with non-linear models.

The non-linear modelling used here was heavily inspired by this tutorial by Ping Hei Yeung.

```
library(aphantasiaReasoningViie)
#> Welcome to aphantasiaReasoningViie.
```

Data preparation

First, let's get the clean, analysis-ready data, create cognitive style clusters using OSIVQ scores (see `vignette("preparing_data")` and `vignette("osivq_clusters")` for details), add this classification to the dataset containing response times (RT) data and filter trials based on RTs (see `vignette("analysing_rt")` for details).

```
df_survey <- get_clean_data()$df_survey
# Clustering OSIVQ data
clustering <- cluster_osivq(df_survey)
# Adding named clusters to the survey data
df_survey <- add_named_clusters(
  df_survey, clustering,
  names = c("Spatialiser", "Visualiser", "Verbaliser"),
  levels = c("Visualiser", "Spatialiser", "Verbaliser"),
  contrasts = c("_visualiser", "_spatialiser", "_verbaliser"),
  base = 1
)
# Merging with experiment data
df_expe <-
  dplyr::left_join(
    get_clean_data()$df_expe,
    df_survey |> dplyr::select(id, cluster),
    by = dplyr::join_by("id")
  ) |>
  dplyr::relocate(cluster, .after = "group")

df_rt <- filter_trials_on_rt(df_expe) |> dplyr::select(id:conclusion_rt)

dplyr::glimpse(df_rt)
#> Rows: 1,904
#> Columns: 15
#> $ id          <fct> acdn247721443631359lzb, acdn247721443631359lzb, acdn2~
```

```
#> $ language      <fct> fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, ~
#> $ group          <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ cluster        <fct> Visualiser, Visualiser, Visualiser, Visualiser, Visuali~
#> $ group_2        <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ group_3        <fct> Typical, Typical, Typical, Typical, Typical, Typical, T~
#> $ strategy_group <fct> No visual strategy, No visual strategy, No visual strat~
#> $ expe_phase     <fct> expe_block_1, expe_block_1, expe_block_1, expe_block_1,~
#> $ trial_number   <int> 2, 5, 7, 8, 9, 11, 14, 15, 16, 17, 18, 19, 20, 21, 22, ~
#> $ problem        <int> 25, 1, 6, 9, 8, 5, 14, 17, 21, 12, 24, 3, 7, 22, 16, 4, ~
#> $ category       <fct> Control, Visual, Visual, Visual, Visual, Visual, Spatia~
#> $ premise_1_rt   <dbl> 7.412, 5.773, 5.596, 4.378, 8.030, 3.038, 4.623, 6.415, ~
#> $ premise_2_rt   <dbl> 3.148, 1.559, 1.646, 7.822, 4.926, 2.881, 4.603, 2.603, ~
#> $ premise_3_rt   <dbl> 4.477, 9.679, 9.206, 3.674, 9.846, 8.729, 8.430, 5.320, ~
#> $ conclusion_rt  <dbl> 3.887, 4.760, 5.369, 4.631, 4.367, 7.507, 8.844, 2.512, ~
```

By default, the `df_rt` data frame has one variable (one column) containing the response times of each participant for each trial term (or “phase”, as referred to in the manuscript — “term” is the word commonly used in VIIIE literature). We needed to have all of these terms (premises and conclusion) in a single “term” variable along with the associated “rt” variable for modelling, so we gathered these four columns into two long columns by “pivoting” the RT data in a long format (reducing the number of columns and increasing the number of rows). This operation is performed by the `pivot_terms_longer()` function.

We also needed to create “interaction” variables between the grouping variables and the categories of the problems (visual, spatial, control) using the `base::interaction()` function because *mgcv* models used for non-linear modelling do not support interaction terms natively.

```
df_rt_long <-
  pivot_terms_longer(df_rt) |>
  dplyr::mutate(
    group_2_category = interaction(group_2, category),
    group_3_category = interaction(group_3, category),
    cluster_category = interaction(cluster, category)
  )

dplyr::glimpse(df_rt_long)
#> Rows: 7,616
#> Columns: 17
#> $ id              <fct> acdn247721443631359lzb, acdn247721443631359lzb, acd~
#> $ language        <fct> fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, fr, f~
#> $ group           <fct> Typical, Typical, Typical, Typical, Typical, Typical, ~
#> $ cluster         <fct> Visualiser, Visualiser, Visualiser, Visualiser, Visua~
#> $ group_2         <fct> Typical, Typical, Typical, Typical, Typical, Typical, ~
#> $ group_3         <fct> Typical, Typical, Typical, Typical, Typical, Typical, ~
#> $ strategy_group  <fct> No visual strategy, No visual strategy, No visual str~
#> $ expe_phase      <fct> expe_block_1, expe_block_1, expe_block_1, expe_block_~
#> $ trial_number    <int> 2, 2, 2, 2, 5, 5, 5, 5, 7, 7, 7, 7, 8, 8, 8, 8, 9, 9, ~
#> $ problem         <fct> 25, 25, 25, 25, 1, 1, 1, 1, 6, 6, 6, 6, 9, 9, 9, 9, 8~
#> $ category        <fct> Control, Control, Control, Control, Visual, Visual, V~
#> $ term            <dbl> 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, ~
#> $ term_name       <fct> Premise 1, Premise 2, Premise 3, Conclusion, Premise ~
#> $ rt              <dbl> 7.412, 3.148, 4.477, 3.887, 5.773, 1.559, 9.679, 4.76~
#> $ group_2_category <fct> Typical.Control, Typical.Control, Typical.Control, Ty~
#> $ group_3_category <fct> Typical.Control, Typical.Control, Typical.Control, Ty~
#> $ cluster_category <fct> Visualiser.Control, Visualiser.Control, Visualiser.Co~
```

Method

As described in the manuscript:

We fitted generalised additive models using the `mgcv` package (Wood, 2011) to predict response times with a grouping variable (VVIQ groups, OSIVQ clusters), Category (visual, spatial or control) and their interaction as fixed categorical predictors, as well as “smooth” (non-linear) terms that capture the evolution of RTs across trial terms (premise 1/2/3 and conclusion) for each grouping and category, each problem for each grouping, and each participant for each category (the latter two being equivalent to “random effects” in mixed models).

Let’s break this down.

The grouping variables

We used several grouping variables to classify participants, all of which are in the `df_rt_long` data frame:

- `group` is the 4-group VVIQ classification with an “aphantasia” group (VVIQ = 16), “hypophantasia” group (VVIQ ∈ [17, 32]), “typical” group (VVIQ ∈ [33, 74]), and “hyperphantasia” group (VVIQ ∈ [75, 80]). It was not used in the analyses because the hyperphantasia group was too small (N = 4).
- `group_3` is the same as `group`, but with the hyperphantasia group merged with the typical group. It is referred to as “VVIQ 3 groups” in the manuscript results.
- `group_2` is the binary classification often used in aphantasia literature that additionally merges the hypophantasia group with the aphantasia group. It is referred to as “VVIQ 2 groups” in the manuscript results.
- `cluster` is the cognitive style classification obtained from clustering OSIVQ scores. It contains three groups: “Visualiser”, “Spatialiser”, and “Verbaliser”. It is referred to as “OSIVQ 3 clusters” in the manuscript results.

The same modelling pipeline was therefore applied three times, once for each of the last three grouping variables.

The modelling pipeline

The models were fitted using the `mgcv::bam()` function. After model fit, we tested group and category differences with marginal contrasts. This task was performed with the `get_contrast()` function, which is a wrapper around several functions from the *emmeans* package.

These models took a long time to fit and needed to be saved for ease of reuse, but model fits were too heavy to be kept as package data. Thus, two sub-products of the models were saved as package data instead:

- Contrast analyses obtained with `get_contrast()` for the statistical tests. The analyses for the three grouping variables have been saved in the `nl_contrasts` list.
- Model predictions obtained with `modelbased::estimate_relation()` for the data used for visualisation. The predictions for the three grouping variables have been saved in the `nl_predictions` list.

The code to fit the models and get the contrasts is provided below but does not run.

Results

VVIQ 2 groups

```
m_nl_vviq_2 <-  
  mgcv::bam(  
    formula = rt ~
```

```

    group_2_category +
    s(term, by = group_2_category, bs = "tp", k = 4) +
    s(term, problem, by = group_2, bs = "fs", m = 1, k = 4) +
    s(term, id, by = category, bs = "fs", m = 1, k = 4),
  family = Gamma(link = "identity"),
  data = df_rt_long,
  method = "fREML"
)

contrasts_vviq_2 <-
  m_nl_vviq_2 |>
  get_contrast(
    ~ group_2_category | term,
    at = list(term = c(1, 2, 3, 4)),
    interaction = FALSE,
    adjust = "none"
  )

```

The chunk above does not run, but the contrasts can be accessed natively in the package in the `nl_contrasts` object. The code below tweaks them a bit for a prettier display:

```

nl_contrasts$vviq_2 |>
  as.data.frame() |>
  tidyr::separate_wider_delim(
    contrast, " - ", names = c("group_cat_1", "group_cat_2")
  ) |>
  tidyr::separate_wider_delim(
    group_cat_1, ".", names = c("group_1", "category_1")
  ) |>
  tidyr::separate_wider_delim(
    group_cat_2, ".", names = c("group_2", "category_2")
  ) |>
  dplyr::filter(group_1 == group_2 & p.value < 1) |>
  dplyr::select(!c(
    tidyselect::contains("group_2"),
    tidyselect::contains("cluster_2"),
    tidyselect::contains("SE"),
    tidyselect::contains("df")
  )) |>
  dplyr::mutate(across(c(estimate:p.value), ~round(., 3))) |>
  tidyr::unite(`Category contrast`, category_1, category_2, sep = " - ") |>
  dplyr::rename(group = group_1, `RT difference` = estimate) |>
  dplyr::arrange(term, group) |>
  dplyr::mutate(
    term = term |>
      as.character() |>
      dplyr::case_match(
        "1" ~ "Premise 1",
        "2" ~ "Premise 2",
        "3" ~ "Premise 3",
        "4" ~ "Conclusion"
      ),
  ) |>
  knitr::kable(digits = 3)

```

```
#> Le chargement a nécessité le package : emmeans
```

group	Category contrast	term	RT difference	t.ratio	p.value
Aphantasia	Control - Spatial	Premise 1	0.239	0.495	0.621
Aphantasia	Control - Visual	Premise 1	-0.095	-0.185	0.853
Aphantasia	Spatial - Visual	Premise 1	-0.334	-0.668	0.504
Typical	Control - Spatial	Premise 1	0.028	0.061	0.951
Typical	Control - Visual	Premise 1	-0.656	-1.323	0.186
Typical	Spatial - Visual	Premise 1	-0.685	-1.409	0.159
Aphantasia	Control - Spatial	Premise 2	-0.681	-1.707	0.088
Aphantasia	Control - Visual	Premise 2	-0.015	-0.037	0.971
Aphantasia	Spatial - Visual	Premise 2	0.666	1.610	0.107
Typical	Control - Spatial	Premise 2	-1.177	-3.070	0.002
Typical	Control - Visual	Premise 2	-0.573	-1.448	0.148
Typical	Spatial - Visual	Premise 2	0.604	1.511	0.131
Aphantasia	Control - Spatial	Premise 3	0.561	1.341	0.180
Aphantasia	Control - Visual	Premise 3	-0.765	-1.704	0.088
Aphantasia	Spatial - Visual	Premise 3	-1.327	-3.015	0.003
Typical	Control - Spatial	Premise 3	0.850	2.104	0.035
Typical	Control - Visual	Premise 3	-0.586	-1.354	0.176
Typical	Spatial - Visual	Premise 3	-1.436	-3.398	0.001
Aphantasia	Control - Spatial	Conclusion	-0.069	-0.152	0.879
Aphantasia	Control - Visual	Conclusion	-0.598	-1.226	0.220
Aphantasia	Spatial - Visual	Conclusion	-0.529	-1.113	0.266
Typical	Control - Spatial	Conclusion	-0.072	-0.164	0.870
Typical	Control - Visual	Conclusion	-0.472	-1.004	0.315
Typical	Spatial - Visual	Conclusion	-0.400	-0.869	0.385

Let's do the same for the other two classifications.

VVIQ 3 groups

```
m_nl_vvqi3_3 <-
  mgcv::bam(
    formula = rt ~
      group_3_category +
      s(term, by = group_3_category, bs = "tp", k = 4) +
      s(term, problem, by = group_3, bs = "fs", m = 1, k = 4) +
      s(term, id, by = category, bs = "fs", m = 1, k = 4),
    family = Gamma(link = "identity"),
    data = df_rt_long,
    method = "fREML"
  )

contrasts_vvqi3_3 <-
  m_nl_vvqi3_3 |>
  get_contrast(
    ~ group_3_category | term,
    at = list(term = c(1, 2, 3, 4)),
    interaction = FALSE,
    adjust = "none"
  )
```

Same process as above, contrasts are saved in the package:

```
nl_contrasts$vvviq_3 |>
  as.data.frame() |>
  tidyr::separate_wider_delim(
    contrast, " - ", names = c("group_cat_1", "group_cat_2")
  ) |>
  tidyr::separate_wider_delim(
    group_cat_1, ".", names = c("group_1", "category_1")
  ) |>
  tidyr::separate_wider_delim(
    group_cat_2, ".", names = c("group_2", "category_2")
  ) |>
  dplyr::filter(group_1 == group_2 & p.value < 1) |>
  dplyr::select(!c(
    tidyselect::contains("group_2"),
    tidyselect::contains("cluster_2"),
    tidyselect::contains("SE"),
    tidyselect::contains("df")
  )) |>
  dplyr::mutate(across(c(estimate:p.value), ~round(., 3))) |>
  tidyr::unite(`Category contrast`, category_1, category_2, sep = " - ") |>
  dplyr::rename(group = group_1, `RT difference` = estimate) |>
  dplyr::arrange(term, group) |>
  dplyr::mutate(
    term = term |>
      as.character() |>
      dplyr::case_match(
        "1" ~ "Premise 1",
        "2" ~ "Premise 2",
        "3" ~ "Premise 3",
        "4" ~ "Conclusion"
      ),
  ) |>
  knitr::kable(digits = 3)
```

group	Category contrast	term	RT difference	t.ratio	p.value
Aphantasia	Control - Spatial	Premise 1	0.464	0.782	0.434
Aphantasia	Control - Visual	Premise 1	0.135	0.212	0.832
Aphantasia	Spatial - Visual	Premise 1	-0.329	-0.534	0.593
Hypophantasia	Control - Spatial	Premise 1	-0.058	-0.083	0.934
Hypophantasia	Control - Visual	Premise 1	-0.494	-0.645	0.519
Hypophantasia	Spatial - Visual	Premise 1	-0.436	-0.587	0.557
Typical	Control - Spatial	Premise 1	0.029	0.063	0.950
Typical	Control - Visual	Premise 1	-0.659	-1.331	0.183
Typical	Spatial - Visual	Premise 1	-0.688	-1.418	0.156
Aphantasia	Control - Spatial	Premise 2	-0.984	-2.026	0.043
Aphantasia	Control - Visual	Premise 2	-0.099	-0.197	0.844
Aphantasia	Spatial - Visual	Premise 2	0.885	1.745	0.081
Hypophantasia	Control - Spatial	Premise 2	-0.328	-0.554	0.580
Hypophantasia	Control - Visual	Premise 2	0.097	0.155	0.877
Hypophantasia	Spatial - Visual	Premise 2	0.425	0.685	0.493
Typical	Control - Spatial	Premise 2	-1.176	-3.088	0.002
Typical	Control - Visual	Premise 2	-0.574	-1.456	0.145

group	Category contrast	term	RT difference	t.ratio	p.value
Typical	Spatial - Visual	Premise 2	0.601	1.508	0.132
Aphantasia	Control - Spatial	Premise 3	0.435	0.842	0.400
Aphantasia	Control - Visual	Premise 3	-0.828	-1.486	0.137
Aphantasia	Spatial - Visual	Premise 3	-1.263	-2.313	0.021
Hypophantasia	Control - Spatial	Premise 3	0.785	1.287	0.198
Hypophantasia	Control - Visual	Premise 3	-0.648	-0.961	0.337
Hypophantasia	Spatial - Visual	Premise 3	-1.434	-2.202	0.028
Typical	Control - Spatial	Premise 3	0.851	2.120	0.034
Typical	Control - Visual	Premise 3	-0.587	-1.358	0.175
Typical	Spatial - Visual	Premise 3	-1.438	-3.412	0.001
Aphantasia	Control - Spatial	Conclusion	-0.159	-0.295	0.768
Aphantasia	Control - Visual	Conclusion	-0.535	-0.913	0.361
Aphantasia	Spatial - Visual	Conclusion	-0.376	-0.657	0.511
Hypophantasia	Control - Spatial	Conclusion	0.111	0.164	0.869
Hypophantasia	Control - Visual	Conclusion	-0.682	-0.910	0.363
Hypophantasia	Spatial - Visual	Conclusion	-0.794	-1.097	0.273
Typical	Control - Spatial	Conclusion	-0.070	-0.160	0.873
Typical	Control - Visual	Conclusion	-0.473	-1.009	0.313
Typical	Spatial - Visual	Conclusion	-0.403	-0.877	0.381

OSIVQ 3 clusters

```

m_nl_osivq <-
  mgcv::bam(
    formula = rt ~
      cluster_category +
      s(term, by = cluster_category, bs = "tp", k = 4) +
      s(term, problem, by = cluster, bs = "fs", m = 1, k = 4) +
      s(term, id, by = category, bs = "fs", m = 1, k = 4),
    family = Gamma(link = "identity"),
    data = df_rt_long,
    method = "fREML"
  )

contrasts_osivq <-
  m_nl_osivq |>
  get_contrast(
    ~ cluster_category | term,
    at = list(term = c(1, 2, 3, 4)),
    interaction = TRUE
  )

```

Same process as above, contrasts are saved in the package:

```

nl_contrasts$osivq |>
  as.data.frame() |>
  tidyr::separate_wider_delim(
    cluster_category_pairwise, " - ", names = c("cluster_cat_1", "cluster_cat_2")
  ) |>
  tidyr::separate_wider_delim(
    cluster_cat_1, ".", names = c("cluster_1", "category_1")
  ) |>

```

```

tidyr::separate_wider_delim(
  cluster_cat_2, ".", names = c("cluster_2", "category_2")
) |>
dplyr::filter(cluster_1 == cluster_2 & p.value < 1) |>
dplyr::select(!c(
  tidyselect::contains("group_2"),
  tidyselect::contains("cluster_2"),
  tidyselect::contains("SE"),
  tidyselect::contains("df")
)) |>
dplyr::mutate(across(c(estimate:p.value), ~round(., 3))) |>
tidyr::unite(`Category contrast`, category_1, category_2, sep = " - ") |>
dplyr::rename(cluster = cluster_1, `RT difference` = estimate) |>
dplyr::arrange(term, cluster) |>
dplyr::mutate(
  term = dplyr::case_match(
    term,
    1 ~ "Premise 1",
    2 ~ "Premise 2",
    3 ~ "Premise 3",
    4 ~ "Conclusion"
  ),
) |>
knitr::kable(digits = 3)

```

cluster	Category contrast	term	RT difference	t.ratio	p.value
Spatialiser	Control - Spatial	Premise 1	0.157	0.229	0.819
Spatialiser	Control - Visual	Premise 1	-0.297	-0.399	0.690
Spatialiser	Spatial - Visual	Premise 1	-0.454	-0.633	0.527
Verbaliser	Control - Spatial	Premise 1	0.447	0.867	0.386
Verbaliser	Control - Visual	Premise 1	0.164	0.299	0.765
Verbaliser	Spatial - Visual	Premise 1	-0.283	-0.533	0.594
Visualiser	Control - Spatial	Premise 1	-0.158	-0.301	0.763
Visualiser	Control - Visual	Premise 1	-0.993	-1.766	0.077
Visualiser	Spatial - Visual	Premise 1	-0.835	-1.514	0.130
Spatialiser	Control - Spatial	Premise 2	-0.681	-1.172	0.241
Spatialiser	Control - Visual	Premise 2	-0.435	-0.714	0.475
Spatialiser	Spatial - Visual	Premise 2	0.245	0.402	0.688
Verbaliser	Control - Spatial	Premise 2	-0.786	-1.862	0.063
Verbaliser	Control - Visual	Premise 2	-0.012	-0.028	0.978
Verbaliser	Spatial - Visual	Premise 2	0.774	1.769	0.077
Visualiser	Control - Spatial	Premise 2	-1.288	-3.001	0.003
Visualiser	Control - Visual	Premise 2	-0.562	-1.267	0.205
Visualiser	Spatial - Visual	Premise 2	0.726	1.623	0.105
Spatialiser	Control - Spatial	Premise 3	0.600	0.983	0.326
Spatialiser	Control - Visual	Premise 3	-0.346	-0.528	0.597
Spatialiser	Spatial - Visual	Premise 3	-0.946	-1.483	0.138
Verbaliser	Control - Spatial	Premise 3	0.539	1.215	0.224
Verbaliser	Control - Visual	Premise 3	-0.785	-1.649	0.099
Verbaliser	Spatial - Visual	Premise 3	-1.324	-2.834	0.005
Visualiser	Control - Spatial	Premise 3	1.003	2.212	0.027
Visualiser	Control - Visual	Premise 3	-0.693	-1.409	0.159
Visualiser	Spatial - Visual	Premise 3	-1.696	-3.565	0.000

cluster	Category contrast	term	RT difference	t.ratio	p.value
Spatialiser	Control - Spatial	Conclusion	0.069	0.105	0.916
Spatialiser	Control - Visual	Conclusion	-0.406	-0.571	0.568
Spatialiser	Spatial - Visual	Conclusion	-0.475	-0.691	0.490
Verbaliser	Control - Spatial	Conclusion	-0.008	-0.016	0.987
Verbaliser	Control - Visual	Conclusion	-0.657	-1.269	0.205
Verbaliser	Spatial - Visual	Conclusion	-0.649	-1.287	0.198
Visualiser	Control - Spatial	Conclusion	-0.189	-0.380	0.704
Visualiser	Control - Visual	Conclusion	-0.478	-0.906	0.365
Visualiser	Spatial - Visual	Conclusion	-0.289	-0.560	0.576

Visualisation

The non-linear dynamics were represented using model predictions for each term, grouping and category. These predictions have been computed with the `modelbased::estimate_relation()` function and the code below (which does not run here):

```
# Getting model predictions (for plotting)
preds_2 <-
  modelbased::estimate_relation(
    m_nl_vviq_2,
    by = c("group_2_category", "term")
  ) |>
  as.data.frame() |>
  dplyr::select(group_2_category, term, Predicted, CI_low, CI_high) |>
  tidyr::separate_wider_delim(
    group_2_category,
    delim = ".",
    names = c("group", "category")
  )

preds_3 <-
  modelbased::estimate_relation(
    m_nl_vviq_3,
    by = c("group_3_category", "term")
  ) |>
  as.data.frame() |>
  dplyr::select(group_3_category, term, Predicted, CI_low, CI_high) |>
  tidyr::separate_wider_delim(
    group_3_category,
    delim = ".",
    names = c("group", "category")
  )

preds_osivq <-
  modelbased::estimate_relation(
    m_nl_osivq,
    by = c("cluster_category", "term")
  ) |>
  as.data.frame() |>
  dplyr::select(cluster_category, term, Predicted, CI_low, CI_high) |>
  tidyr::separate_wider_delim(
    cluster_category,
```

```

delim = ".",
names = c("group", "category")
)

```

... And have been saved as package data in the `nl_predictions` object. We created a `plot_nl()` function to visualise the data easily and added significance labels based on the contrasts with the `add_significance()` helper function (which is why the code below is so lengthy).

```

library(patchwork)

pn11 <-
  plot_nl(
    nl_predictions$vv1q_2, title = "VVIQ 2 groups",
    plot.background = ggplot2::element_rect(fill = "white", colour = NA)
  ) +

  # 2nd premise
  add_significance(
    size_star = 3,
    tibble::tibble(
      group = factor(c("Aphantasia")),
      x_star = 1.93,
      y_star = 5.97,
      stars = "°",
      x_line = .data$x_star - 0.06,
      x_line_end = .data$x_star + 0.06,
      y_line = 5.9
    )
  ) +
  add_significance(
    size_star = 3,
    tibble::tibble(
      group = factor(c("Typical")),
      x_star = 1.93,
      y_star = 6.24,
      stars = "***",
      x_line = .data$x_star - 0.06,
      x_line_end = .data$x_star + 0.06,
      y_line = 6.17
    )
  ) +

  # 3rd premise
  add_significance(
    size_star = 3,
    tibble::tibble(
      group = factor(c("Aphantasia")),
      x_star = 3.06,
      y_star = 7.87,
      stars = "***",
      x_line = .data$x_star - 0.05,
      x_line_end = .data$x_star + 0.05,
      y_line = 7.8
    )
  )

```

```

) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Aphantasia")),
    x_star = 3.01,
    y_star = 8.27,
    stars = "°",
    x_line = .data$x_star - 0.1,
    x_line_end = .data$x_star + 0.1,
    y_line = 8.2
  )
) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Typical")),
    x_star = 3.06,
    y_star = 8.47,
    stars = "***",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,
    y_line = 8.4
  )
) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Typical")),
    x_star = 2.93,
    y_star = 7.77,
    stars = "*",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,
    y_line = 7.7
  )
)
)

pn12 <-
plot_n1(
  nl_predictions$vv1q_3,
  title = "VVIQ 3 groups",
  plot.margin = ggplot2::margin(t = 10),
  plot.background = ggplot2::element_rect(fill = "white", colour = NA)
) +

# 2nd premise
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Aphantasia")),
    x_star = 1.93,
    y_star = 6.63,

```

```

    stars = "*",
    x_line = .data$x_star - 0.06,
    x_line_end = .data$x_star + 0.06,
    y_line = 6.57
  )
) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Typical")),
    x_star = 1.93,
    y_star = 6.24,
    stars = "**",
    x_line = .data$x_star - 0.06,
    x_line_end = .data$x_star + 0.06,
    y_line = 6.17
  )
) +

# 3rd premise
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Aphantasia")),
    x_star = 3.06,
    y_star = 8.32,
    stars = "*",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,
    y_line = 8.25
  )
) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Hypophantasia")),
    x_star = 3.06,
    y_star = 7.62,
    stars = "*",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,
    y_line = 7.55
  )
) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Typical")),
    x_star = 3.06,
    y_star = 8.47,
    stars = "***",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,

```

```

    y_line = 8.4
  )
) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Typical")),
    x_star = 2.93,
    y_star = 7.77,
    stars = "*",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,
    y_line = 7.7
  )
)

pn13 <-
plot_n1(
  nl_predictions$osivq,
  title = "OSIVQ clusters",
  plot.margin = ggplot2::margin(t = 10),
  plot.background = ggplot2::element_rect(fill = "white", colour = NA)
) +

# 1st premise
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Visualiser")),
    x_star = 1,
    y_star = 8.47,
    stars = "o",
    x_line = .data$x_star - 0.1,
    x_line_end = .data$x_star + 0.1,
    y_line = 8.4
  )
) +

# 2nd premise
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Visualiser")),
    x_star = 1.93,
    y_star = 6.37,
    stars = "***",
    x_line = .data$x_star - 0.06,
    x_line_end = .data$x_star + 0.06,
    y_line = 6.3
  )
) +
add_significance(
  size_star = 3,

```

```

tibble::tibble(
  group = factor(c("Verbaliser")),
  x_star = 1.93,
  y_star = 6.17,
  stars = "o",
  x_line = .data$x_star - 0.06,
  x_line_end = .data$x_star + 0.06,
  y_line = 6.1
)
) +

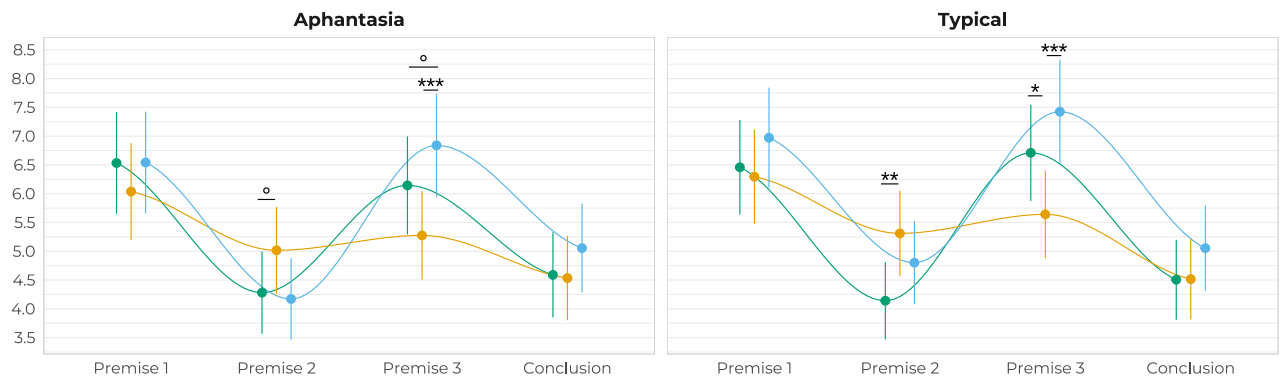
# 3rd premise
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Visualiser")),
    x_star = 3.06,
    y_star = 8.62,
    stars = "***",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,
    y_line = 8.55
  )
) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Visualiser")),
    x_star = 2.93,
    y_star = 7.77,
    stars = "*",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,
    y_line = 7.7
  )
) +
add_significance(
  size_star = 3,
  tibble::tibble(
    group = factor(c("Verbaliser")),
    x_star = 3.06,
    y_star = 8.04,
    stars = "*",
    x_line = .data$x_star - 0.05,
    x_line_end = .data$x_star + 0.05,
    y_line = 7.97
  )
)

pnl <- pnl1 / pnl2 / pnl3 +
  patchwork::plot_layout(axes = "collect", guides = "collect") &
  ggplot2::theme(legend.position = "bottom")

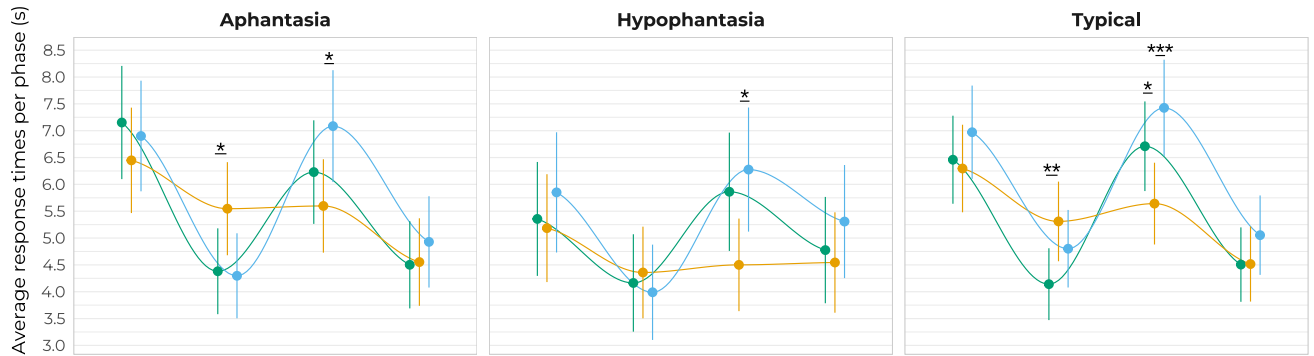
```

```
plot(pnl)
```

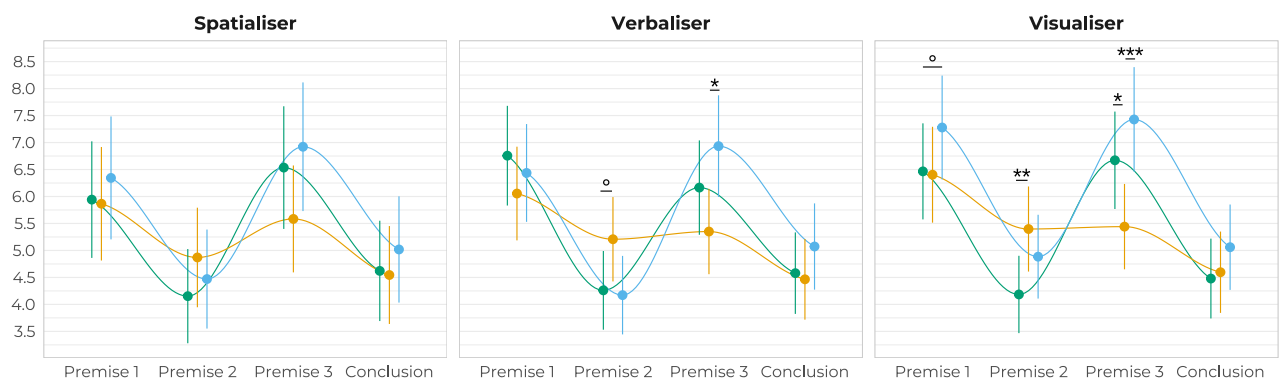
VVIQ 2 groups



VVIQ 3 groups



OSIVQ clusters



Problem category: ◆ Control ◆ Spatial ◆ Visual

```

#> - Session info -----
#> setting value
#> version R version 4.5.0 (2025-04-11 ucrt)
#> os      Windows 11 x64 (build 26100)
#> system  x86_64, mingw32
#> ui      RTerm
#> language (EN)
#> collate French_France.utf8
#> ctype   French_France.utf8
#> tz      Europe/Paris
#> date    2025-09-17
#> pandoc  3.6.3 @ C:/Program Files/RStudio/resources/app/bin/quarto/bin/tools/ (via rmarkdown)
#>
#> - Packages -----
#> ! package          * version      date (UTC) lib source
#> P abind             1.4-8        2024-09-12 [?] CRAN (R 4.5.0)
#> P aphantasiaReasoningViie * 0.0.0.9000 2025-09-16 [1] local
#> P assertthat        0.2.1        2019-03-21 [?] CRAN (R 4.5.0)
#> P class             7.3-23       2025-01-01 [3] CRAN (R 4.5.0)
#> P cli               3.6.5        2025-04-23 [?] CRAN (R 4.5.0)
#> P clue              0.3-66       2024-11-13 [?] CRAN (R 4.5.0)
#> P cluster           2.1.8.1      2025-03-12 [3] CRAN (R 4.5.0)
#> P clusterCrit       1.3.0        2023-11-23 [?] CRAN (R 4.5.0)
#> P clValid           0.7          2021-02-14 [?] CRAN (R 4.5.0)
#> P coda              0.19-4.1     2024-01-31 [?] CRAN (R 4.5.0)
#> P combinat          0.0-8        2012-10-29 [?] CRAN (R 4.5.0)
#> P crayon            1.5.3        2024-06-20 [?] CRAN (R 4.5.0)
#> P curl              6.3.0        2025-06-06 [?] RSPM
#> P diceR             3.1.0        2025-06-19 [?] RSPM (R 4.5.0)
#> P digest            0.6.37       2024-08-19 [?] CRAN (R 4.5.0)
#> P dplyr             1.1.4        2023-11-17 [?] CRAN (R 4.5.0)
#> P e1071             1.7-16       2024-09-16 [?] CRAN (R 4.5.0)
#> P emmeans           1.11.1       2025-05-04 [?] CRAN (R 4.5.0)
#> P estimability      1.5.1        2024-05-12 [?] CRAN (R 4.5.0)
#> P evaluate          1.0.4        2025-06-18 [?] RSPM (R 4.5.0)
#> P farver            2.1.2        2024-05-13 [?] CRAN (R 4.5.0)
#> P fastmap           1.2.0        2024-05-15 [?] CRAN (R 4.5.0)
#> P forcats           1.0.0        2023-01-29 [?] CRAN (R 4.5.0)
#> P generics          0.1.4        2025-05-09 [?] CRAN (R 4.5.0)
#> P ggplot2           3.5.2        2025-04-09 [?] CRAN (R 4.5.0)
#> P glue              1.8.0        2024-09-30 [?] CRAN (R 4.5.0)
#> P gtable            0.3.6        2024-10-25 [?] CRAN (R 4.5.0)
#> P haven             2.5.5        2025-05-30 [?] RSPM (R 4.5.0)
#> P highr             0.11         2024-05-26 [?] CRAN (R 4.5.0)
#> P hms               1.1.3        2023-03-21 [?] CRAN (R 4.5.0)
#> P htmltools         0.5.8.1      2024-04-04 [?] CRAN (R 4.5.0)
#> P httpuv            1.6.16       2025-04-16 [?] CRAN (R 4.5.0)
#> P jsonlite          2.0.0        2025-03-27 [?] CRAN (R 4.5.0)
#> P klaR              1.7-3        2023-12-13 [?] CRAN (R 4.5.0)
#> P knitr             1.50         2025-03-16 [?] CRAN (R 4.5.0)
#> P labelled          2.14.1       2025-05-06 [?] CRAN (R 4.5.0)
#> P later             1.4.2        2025-04-08 [?] CRAN (R 4.5.0)
#> P lattice           0.22-7       2025-04-02 [?] CRAN (R 4.5.0)
#> P lifecycle         1.0.4        2023-11-07 [?] CRAN (R 4.5.0)

```

```

#> P magrittr          2.0.3      2022-03-30 [?] CRAN (R 4.5.0)
#> P MASS              7.3-65     2025-02-28 [3] CRAN (R 4.5.0)
#> P Matrix            1.7-3      2025-03-11 [3] CRAN (R 4.5.0)
#> P mclust            6.1.1      2024-04-29 [?] CRAN (R 4.5.0)
#> P mgcv              1.9-3      2025-04-04 [?] CRAN (R 4.5.0)
#> P mime              0.13       2025-03-17 [?] CRAN (R 4.5.0)
#> P miniUI            0.1.2      2025-04-17 [?] CRAN (R 4.5.0)
#> P mvtnorm           1.3-3      2025-01-10 [?] CRAN (R 4.5.0)
#> P nlme              3.1-168    2025-03-31 [3] CRAN (R 4.5.0)
#> P patchwork         * 1.3.1     2025-06-21 [?] RSPM (R 4.5.0)
#> P pillar            1.10.2     2025-04-05 [?] CRAN (R 4.5.0)
#> P pkgconfig         2.0.3      2019-09-22 [?] CRAN (R 4.5.0)
#> P promises          1.3.3      2025-05-29 [?] RSPM
#> P proxy             0.4-27     2022-06-09 [?] CRAN (R 4.5.0)
#> P purrr             1.0.4      2025-02-05 [?] CRAN (R 4.5.0)
#> P questionr        0.8.1      2025-06-10 [?] RSPM (R 4.5.0)
#> P R6                2.6.1      2025-02-15 [?] CRAN (R 4.5.0)
#> P RColorBrewer      1.1-3      2022-04-03 [?] CRAN (R 4.5.0)
#> P Rcpp              1.0.14     2025-01-12 [?] CRAN (R 4.5.0)
#> P rlang             1.1.6      2025-04-11 [?] CRAN (R 4.5.0)
#> P rmarkdown         2.29       2024-11-04 [?] CRAN (R 4.5.0)
#> P rstudioapi        0.17.1     2024-10-22 [?] CRAN (R 4.5.0)
#> P scales            1.4.0      2025-04-24 [?] CRAN (R 4.5.0)
#> P sessioninfo       1.2.2      2021-12-06 [?] RSPM (R 4.5.0)
#> P shiny             1.10.0     2024-12-14 [?] CRAN (R 4.5.0)
#> P showtext          0.9-7      2024-03-02 [?] CRAN (R 4.5.0)
#> P showtextdb        3.0        2020-06-04 [?] CRAN (R 4.5.0)
#> P stringi           1.8.7      2025-03-27 [?] CRAN (R 4.5.0)
#> P stringr           1.5.1      2023-11-14 [?] CRAN (R 4.5.0)
#> P sysfonts          0.8.9      2024-03-02 [?] CRAN (R 4.5.0)
#> P tibble            3.3.0      2025-06-08 [?] RSPM
#> P tidyr             1.3.1      2024-01-24 [?] CRAN (R 4.5.0)
#> P tidyselect        1.2.1      2024-03-11 [?] CRAN (R 4.5.0)
#> P vctrs             0.6.5      2023-12-01 [?] CRAN (R 4.5.0)
#> P withr             3.0.2      2024-10-28 [?] CRAN (R 4.5.0)
#> P xfun              0.52       2025-04-02 [?] CRAN (R 4.5.0)
#> P xtable            1.8-4      2019-04-21 [?] CRAN (R 4.5.0)
#> P yaml              2.3.10     2024-07-26 [?] CRAN (R 4.5.0)
#>
#> [1] C:/Users/Mael/AppData/Local/R/cache/R/renv/library/aphantasiaReasoningViie-d318fbb0/windows/R-4
#> [2] C:/Users/Mael/AppData/Local/R/cache/R/renv/sandbox/windows/R-4.5/x86_64-w64-mingw32/9bc714e4
#> [3] C:/Program Files/R/R-4.5.0/library
#>
#> P -- Loaded and on-disk path mismatch.
#>
#> -----

```